

Couette Shear on the Infinite Periodic Channel: Exact Fourier Semigroup, Sharp Enhanced Dissipation, and Every Boundary

Route-A structural certificate C206

Abstract

For every real shear a , viscosity $\nu \geq 0$, and physical time, we derive the complete Fourier semigroup of scalar Couette advection–diffusion on $\mathbb{T} \times \mathbb{R}$. Completing one exact quadratic integral gives the exact and sharp k -sector norm $\exp\{-\nu(k^2t + a^2k^2t^3/12)\}$ and its cubic enhanced-dissipation scale, together with the precise norm-attainment boundary. We prove composition, inviscid unitary mixing, all no-shear/zero-mode/zero-time boundaries, the full L^2 periodic-state classification, and noncompactness excluding an ordinary Fredholm determinant. Finite exact cells audit conventions but do not prove the continuous theorem. The inviscid unitary is only a formal lift: all arithmetic and determinant gates fail, so Route A is rejected.

Keywords: Couette flow; shear mixing; enhanced dissipation; Fourier semigroup; noncompactness.

中文摘要

本文对无限周期通道上的Couette对流扩散方程给出全部实剪切、非负黏性与物理时间下的精确傅里叶半群。一次平方配方得到精确且尖锐的扇区范数、范数取等边界和三次增强耗散尺度，并统一处理无黏、无剪切、零模、零时刻、周期态及非迹类边界。有限证书只校验约定；由于不存在内生素数载体与目标行列式，路线A仍被拒绝。

1 Frozen equation and Fourier sign

Let $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ and consider

$$\partial_t f + ay\partial_x f = \nu(\partial_x^2 + \partial_y^2)f, \quad (x, y) \in \mathbb{T} \times \mathbb{R}, \quad a \in \mathbb{R}, \quad \nu \geq 0. \quad (1)$$

The clock is physical PDE time. We freeze

$$\widehat{f}_k(\eta) = \frac{1}{2\pi} \int_{\mathbb{T}} \int_{\mathbb{R}} f(x, y) e^{-i(kx + \eta y)} dy dx.$$

Since multiplication by y becomes $i\partial_\eta$, the transformed equation is

$$\partial_t \widehat{f}_k - ak\partial_\eta \widehat{f}_k = -\nu(k^2 + \eta^2)\widehat{f}_k. \quad (2)$$

Couette damping and enhanced-dissipation context is classical [1]; we claim a convention-complete derivation and certificate, not priority.

2 Exact semigroup and composition

Theorem 1. For every $k \in \mathbb{Z}$, $\eta \in \mathbb{R}$, and $t \geq 0$,

$$\widehat{S}_t \widehat{f}_k(\eta) = \exp \left[-\nu \left\{ k^2 t + t \left(\eta + \frac{akt}{2} \right)^2 + \frac{a^2 k^2 t^3}{12} \right\} \right] \widehat{f}_k(\eta + akt). \quad (3)$$

Moreover $S_t S_s = S_{t+s}$ on $L^2(\mathbb{T} \times \mathbb{R})$.

Proof. The characteristic ending at (t, η) is $\eta(r) = \eta + ak(t - r)$. Integrating (2) gives

$$D_t(\eta) = k^2 t + \int_0^t (\eta + akr)^2 dr = k^2 t + \eta^2 t + ak\eta t^2 + \frac{a^2 k^2 t^3}{3}.$$

Completing the square yields (3). Direct expansion gives $D_t(\eta) + D_s(\eta + akt) = D_{t+s}(\eta)$ and the shifts add, proving composition. $\square$

3 Sharp sector norm and scale

Proposition 1. On the k -th Fourier sector,

$$\|S_t\|_{k \rightarrow k} = \exp \left[-\nu \left(k^2 t + \frac{a^2 k^2 t^3}{12} \right) \right]. \quad (4)$$

If $\nu t > 0$, no nonzero L^2 vector attains this norm; frequency-localized packets approach it. If $\nu t = 0$, every nonzero vector attains the norm.

Proof. Write the sector operator as $\mathcal{M}_{m_t}U_t$, where $U_tg(\eta) = g(\eta + akt)$ is unitary and $m_t(\eta)$ is the positive exponential in (3). Its essential supremum is the value in (4): continuity and the unique quadratic minimizer $\eta_0 = -akt/2$ give the upper bound, while normalized functions supported in shrinking intervals about η_0 approach equality.

Let N denote the value in (4). If $\nu t > 0$ and a nonzero $g \in L^2(\mathbb{R}_\eta)$ attained the norm, equality of squared norms would give

$$\int_{\mathbb{R}} (N^2 - |m_t(\eta)|^2) |U_tg(\eta)|^2 d\eta = 0.$$

The first factor is strictly positive away from the singleton $\{\eta_0\}$, so U_tg would be supported on a null set and hence vanish, a contradiction. If $\nu t = 0$, $m_t \equiv 1$ and $\mathcal{M}_{m_t}U_t$ is unitary (the identity when $t = 0$), so every nonzero vector attains its norm. $\square$

For $avk \neq 0$, the cubic factor reaches order one at $t = (12/(\nu a^2 k^2))^{1/3}$; the ordinary heat term $\nu k^2 t$ remains.

4 Inviscid mixing and every degenerate boundary

At $\nu = 0$, (1) is the unitary shear group

$$f_k(y, t) = e^{-iakty} f_k(y, 0).$$

For $f_k \bar{g} \in L^1$, its pairing with g tends to zero for $ak \neq 0$ by the Riemann–Lebesgue lemma, while its L^2 norm is constant. Thus weak mixing is not dissipative decay. The involution $(x, y) \mapsto (-x, y)$ reverses this same clock. Positive viscosity destroys unitarity.

If $a = 0$, (3) is ordinary heat. If $k = 0$, it is one-dimensional heat in y , whose operator norm is one on the noncompact line. If $t = 0$, it is the identity. These statements include the joint $a = \nu = 0$ identity boundary.

5 Periodic states and the trace-class stop

Proposition 2. *Let $T > 0$. If $\nu > 0$ and $S_T f = f$ in L^2 , then $f = 0$. If $\nu = 0$ and $aT \neq 0$, the T -periodic states are exactly the streamwise means.*

Proof. For $k \neq 0$ and $\nu > 0$, (4) is strictly below one. For $k = 0$, the equality $(1 - e^{-\nu T \eta^2}) \hat{f}_0(\eta) = 0$ confines support to the null set $\{0\}$. In the inviscid case, periodicity makes every nonzero k component periodic in η with nonzero period akT ; an $L^2(\mathbb{R})$ periodic function is zero. The $k = 0$ component is fixed. $\square$

For $t > 0$, the $k = 0$ sector is multiplication by $e^{-\nu t \eta^2}$ on the nonatomic space $L^2(\mathbb{R}_\eta)$. A nonzero multiplication operator there is not compact. Hence the full semigroup is noncompact and not trace class (the inviscid unitary and $t = 0$ identity are also noncompact). We therefore define no ordinary Fredholm determinant.

Revision focus. Round 1 adds inviscid mixing, every parameter boundary, all periodic states, reversal, and the non-trace-class theorem.

Declarations

Data and code availability. Package-local exact synthetic evidence and deterministic code accompany the paper; no observational data are used.

Ethics. No human, animal, personal, or sensitive data are used.

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References

- [1] J. Bedrossian, N. Masmoudi, and V. Vicol, “Enhanced dissipation and inviscid damping in the inviscid limit of the Navier–Stokes equations near the two dimensional Couette flow,” *Arch. Rational Mech. Anal.* 219 (2016), 1087–1159. DOI: 10.1007/s00205-015-0917-3.